Ex.No: 06	PL/SQL – FUNCTIONS AND PROCEDURES
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Objective

To create PL/SQL for various operations.

Description

- PL/SQL, which stands for Procedural Language extensions to the Structured Query Language (SQL).
- It is a combination of SQL along with the procedural features of programming languages.
- It was developed by Oracle Corporation in the early 90's to enhance the capabilities of SQL.
- Declare section starts with DECLARE keyword in which variables, constants, records as cursors can be declared which stores data temporarily. It basically consists definition of PL/SQL identifiers. This part of the code is optional.
- Execution section starts with BEGIN and ends with END keyword. This is a mandatory section and here the program logic is written to perform any task like loops and conditional statements. It supports all DML commands, DDL commands and SQL*PLUS built-in functions as well.
- Exception section starts with EXCEPTION keyword. This section is optional which contains statements that are executed when a run-time error occurs. Any exceptions can be handled in this section.
- PL/SQL supports various data types like NUMBER, VARCHAR2, DATE, etc.
- Variables can be declared in the DECLARE section of a block.

Syntax:

```
DECLARE
    declaration statements;

BEGIN
    executable statements

EXCEPTIONS
    exception handling statements

END;
```

Sample Program:

1. simple PL/SQL program that calculates the square of a number and prints the result:

```
DECLARE
message varchar2(100):= 'Hello, World!';
BEGIN
dbms_output.put_line(message);
END;
```

2. simple PL/SQL program that calculates the square of a number and prints the result:

```
DECLARE
  -- Declare a variable to hold the number
  v_number NUMBER := 5;
    -- Declare a variable to hold the result
  v_square NUMBER;

BEGIN
  -- Calculate the square of the number
  v_square := v_number * v_number;
    -- Output the result
  DBMS_OUTPUT.PUT_LINE('The square of ' || v_number || ' is ' || v_square);

END;
```

3. Example of PL/SQL case statement

```
DECLARE
    grade char(1) := 'A';

 BEGIN
    CASE grade
      when 'A' then dbms_output.put_line('Excellent');
      when 'B' then dbms_output.put_line('Very good');
      when 'C' then dbms_output.put_line('Good');
      when 'D' then dbms_output.put_line('Average');
      when 'F' then dbms_output.put_line('Passed with Grace');
      else dbms_output.put_line('Failed');

    END CASE;
  END;
```

Questions:

1. Write a PL/SQL block to print first n natural numbers and their sum.
2. Write a PL/SQL block to find out sum of the digits of a given number
3. Write a PL/SQL block to generate a Fibonacci series
4. Write a PL/SQL script that traps ZERO_DIVIDE exceptions.
5. Write a PL/SQL block to illustrate functioning of IF_THEN_ELSIF by creating the equivalent of a four-function calculator. The program requires the user to enter two numbers and an operator. It then carries out the specified arithmetical operation: addition, subtraction, multiplication or division of the two numbers. Finally, it displays the result.
6. Write a PL/SQL Procedure to find the area of the circle.
7. Write a PL/SQL Function to find the area of the rectangle.

Result: